

Markov Chains

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Introduction

Markov Chains are mathematical models used to describe systems that change over time in a probabilistic manner.

- Systems evolve step-by-step
- Future behavior can be predicted using probabilities

Examples:

- Student performance
- Brand switching
- Machine condition

Relation to Stochastic Processes

A stochastic process is a sequence of random variables:

$$\{X_0, X_1, X_2, \dots\}$$

Markov Chain is a special case where:

- State space is discrete
- Future depends only on present state

Markov Property

$$P(X_n | X_0, \dots, X_{n-1}) = P(X_n | X_{n-1})$$

- Future depends only on present
- Past history is ignored

When to Use Markov Chains

Use when:

- System changes step-by-step
- Only present state affects future
- Transition probabilities are known

Do not use when:

- Past history significantly affects future

Types of Markov Chains

- Discrete-Time Markov Chain (DTMC)
- Continuous-Time Markov Chain (CTMC)
- Homogeneous and Non-homogeneous
- Absorbing Markov Chain

How to Identify Type

- Time in steps $\rightarrow$ DTMC
- Continuous time $\rightarrow$ CTMC
- Constant probabilities $\rightarrow$ Homogeneous
- Changing probabilities $\rightarrow$ Non-homogeneous
- Trapping state $\rightarrow$ Absorbing

Transition Matrix

$$p_{ij} = P(X_{n+1} = j | X_n = i)$$

$$P = \begin{bmatrix} p_{11} & p_{12} \\ p_{21} & p_{22} \end{bmatrix}$$

- Row sum = 1
- All entries between 0 and 1

n-Step Transition (Concept)

$$p_{ij}^{(n)} = P(X_{k+n} = j | X_k = i)$$

- Probability of reaching state j from i in n steps
- Used for predicting future states

Matrix Form of n-Step Transition

$$P^n = P \cdot P \dots P$$

$$\pi^{(n)} = \pi^{(0)} P^n$$

- $P^2 \rightarrow$ 2-step probabilities
- $P^3 \rightarrow$ 3-step probabilities

Chapman-Kolmogorov Equation

$$p_{ij}^{(n)} = \sum_k p_{ik}^{(m)} p_{kj}^{(n-m)}$$

- Break transition into intermediate steps
- Basis of matrix multiplication

Steady-State Distribution

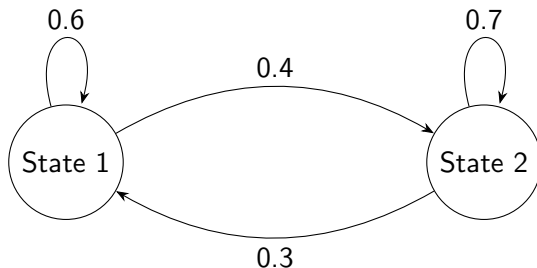
$$\pi P = \pi$$

$$\sum \pi_i = 1$$

- Represents long-run behavior
- System reaches equilibrium

Understanding π using State Diagram

Example: Two-State Markov Chain



Probability Vector:

$$\pi = [\pi_1 \quad \pi_2]$$

- π_1 = probability of being in State 1
- π_2 = probability of being in State 2

Steady-State Meaning:

$$\pi P = \pi$$

- System keeps moving between states
- Long-run proportions remain constant
- π_i = long-run proportion of time in state i
- Independent of initial state (in regular chains)

Meaning of $\pi P = \pi$

Steady-State Equation:

$$\pi P = \pi$$

Interpretation:

- The probability distribution π does not change after transition
- System has reached **equilibrium (steady-state)**

Mathematical Form: Let $\pi = [\pi_1 \ \pi_2]$

$$\pi P = [\pi_1 p_{11} + \pi_2 p_{21} \quad \pi_1 p_{12} + \pi_2 p_{22}]$$

$$\Rightarrow \pi_1 = \pi_1 p_{11} + \pi_2 p_{21}$$

$$\Rightarrow \pi_2 = \pi_1 p_{12} + \pi_2 p_{22}$$

Initial State Vector in Markov Chains

General Form:

$$\pi^{(0)} = [\pi_1^{(0)} \ \pi_2^{(0)}]$$

- $\pi_1^{(0)}$ = Probability of State 1
- $\pi_2^{(0)}$ = Probability of State 2

Case 1: System starts in State 1

$$\pi^{(0)} = [1 \ 0]$$

- 100% in State 1
- 0% in State 2

Case 2: System starts in State 2

$$\pi^{(0)} = [0 \ 1]$$

- 0% in State 1
- 100% in State 2

Important:

- Order must match transition matrix
- Format = [State 1 State 2]

Meaning in Words:

- Long-run probabilities remain constant
- System keeps moving, but proportions do not change

Example 1: Student Performance (Question)

A student can either Pass (P) or Fail (F) in an exam.

Transition information:

- If a student passes, probability of passing next time = 0.6
- If a student fails, probability of passing next time = 0.3

Find:

- 1 Probability that a student who passes today will fail after 2 exams
- 2 State probabilities after 2 steps
- 3 Steady-state distribution

Example 1: Step 1 (Transition Matrix)

States:

- P = Pass
- F = Fail

$$P = \begin{bmatrix} 0.6 & 0.4 \\ 0.3 & 0.7 \end{bmatrix}$$

Explanation:

- Row 1 → current state Pass
- Row 2 → current state Fail

Example 1: Step 2 (Compute P^2)

$$P^2 = P \cdot P$$

$$= \begin{bmatrix} 0.6 & 0.4 \\ 0.3 & 0.7 \end{bmatrix} \begin{bmatrix} 0.6 & 0.4 \\ 0.3 & 0.7 \end{bmatrix}$$

$$p_{11}^{(2)} = 0.6(0.6) + 0.4(0.3) = 0.48$$

$$p_{12}^{(2)} = 0.6(0.4) + 0.4(0.7) = 0.52$$

Example 1: Step 2 (Continued)

$$p_{21}^{(2)} = 0.3(0.6) + 0.7(0.3) = 0.39$$

$$p_{22}^{(2)} = 0.3(0.4) + 0.7(0.7) = 0.61$$

$$P^2 = \begin{bmatrix} 0.48 & 0.52 \\ 0.39 & 0.61 \end{bmatrix}$$

Example 1: Step 3 (Answers)

(1) Probability of Pass $\rightarrow$ Fail in 2 steps:

$$= 0.52$$

(2) State probabilities after 2 steps:

$$\pi^{(0)} = [1 \ 0]$$

$$\pi^{(2)} = [0.48 \ 0.52]$$

Interpretation:

- 48% Pass
- 52% Fail

Example 1: Step 4 (Steady-State)

$$\pi_P = 0.6\pi_P + 0.3\pi_F$$

$$\pi_P + \pi_F = 1$$

$$0.4\pi_P = 0.3\pi_F \Rightarrow \pi_P = \frac{3}{4}\pi_F$$

$$\pi = \begin{bmatrix} \frac{3}{7} & \frac{4}{7} \end{bmatrix}$$

Example 2: Brand Switching (Question)

A customer chooses between Brand A and Brand B.

Transition information:

- If a customer chooses A, probability of staying with A = 0.5
- If a customer chooses B, probability of switching to A = 0.2

Find:

- 1 Probability of $A \rightarrow B$ in 2 steps
- 2 State probabilities after 2 steps
- 3 Steady-state distribution

Example 2: Step 1 (Transition Matrix)

$$P = \begin{bmatrix} 0.5 & 0.5 \\ 0.2 & 0.8 \end{bmatrix}$$

Explanation:

- Row 1 → Brand A
- Row 2 → Brand B

Example 2: Step 2 (Compute P^2)

$$P^2 = \begin{bmatrix} 0.35 & 0.65 \\ 0.26 & 0.74 \end{bmatrix}$$

(Derived using matrix multiplication)

Example 2: Step 3 (Answers)

(1) Probability of $A \rightarrow B$ in 2 steps:

$$= 0.65$$

(2) State probabilities after 2 steps:

$$\pi^{(2)} = [0.35 \quad 0.65]$$

Example 2: Step 4 (Steady-State)

$$\pi_A = 0.5\pi_A + 0.2\pi_B$$

$$\pi_A + \pi_B = 1$$

$$0.5\pi_A = 0.2\pi_B \Rightarrow \pi_A = \frac{2}{5}\pi_B$$

$$\pi = \begin{bmatrix} \frac{2}{7} & \frac{5}{7} \end{bmatrix}$$

Final Summary

- Markov chains model memoryless systems
- Use transition matrix to describe behavior
- Use P^n for future prediction
- Use $\pi P = \pi$ for steady-state